

Albert Gafiyatullin

AI Compiler Engineer

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SUMMARY

Software Engineer specializing in the development of programming language compilers and runtimes, with experience in AI compiler optimizations, JIT compilation, and JVM internals.

EXPERIENCE

Samsung Research Russia

Moscow, Russia

July 2023 - Present

- Leading AI Compiler Engineer, SOC SW Lab

C++ Compilers Neural Networks Optimizations

Developing an optimizing AI compiler for the Samsung Exynos SoC NPU:

- Designed and implemented new generic SPM-allocation, post-allocation, and scheduling optimizations in the NPU compiler, improving neural-network performance and energy efficiency on Samsung mobile devices;
- Reduced inference latency for target neural networks through performance analysis and tuning.

Unipro

Novosibirsk, Russia

March 2021 - June 2023

- JIT Compiler Engineer, Java Virtual Machine Team

C++ JVM Assembly Compilers Garbage Collection

JVM runtime and JIT compiler development for MCST's Elbrus VLIW processor:

- Adapted the C2 compiler to support tiered compilation, improving average startup performance by 50% compared with non-tiered compilation;
- Designed and implemented a fast "C0" compiler for warm-up compilation tiers, replacing C1 and targeting the Elbrus VLIW architecture; improved average startup performance by 25% on low-core-count systems compared with C2-based tiered compilation;
- Improved performance of string- and XML-related workloads by up to 8% using intrinsics;
- Reduced runtime overhead by implementing platform-specific improvements to implicit null checks.

PROJECTS & COURSES

COOL Compiler

- C++ LLVM Garbage Collection Compilers ARM

March 2024

Implemented a COOL compiler and runtime using LLVM:

- Supported AArch64 and x86-64 target architectures;
- Implemented stop-the-world mark-and-sweep, mark-and-compact and semispace copying garbage collectors.

SOE.YCSCS1: Compilers

- C++ MIPS Compilers Assembly

October 2021

Implemented a COOL compiler for the SPIM emulator.

ENGR85A: Digital Design

- SystemVerilog Circuit Design

April 2023

Combinational and sequential circuit design.

ENGR85B: Computer Architecture

- RISC-V Computer Architecture SystemVerilog Embedded Systems Circuit Design

June 2023

Implemented a [multicycle RISC-V CPU](#) and studied introductory pipelined CPU design.

EDUCATION

Novosibirsk State University

Novosibirsk, Russia

- Master's degree in Computer Science

2021 - 2023

- Thesis:** Tiered JIT Compilation in the Java Virtual Machine for the Elbrus Platform.

Novosibirsk State University

Novosibirsk, Russia

- Bachelor's degree in Computer Science

2017 - 2021

- Thesis:** Development of a computational module for simulating the destruction of a fast-neutron reactor core.

- GPA:** 4.8/5.0, graduated with honors

PROGRAMMING SKILLS

- Languages:** C++, C, assembly, Python.
- Technologies:** JVM internals, compiler optimization, JIT compilation, computer architecture, LLVM, garbage collection